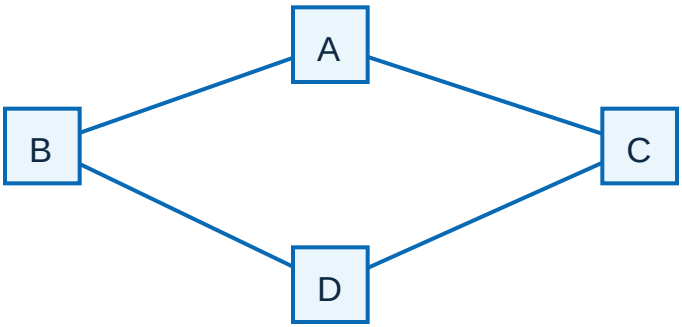
Breadth-first search

Use a queue to visit the graph level by level.



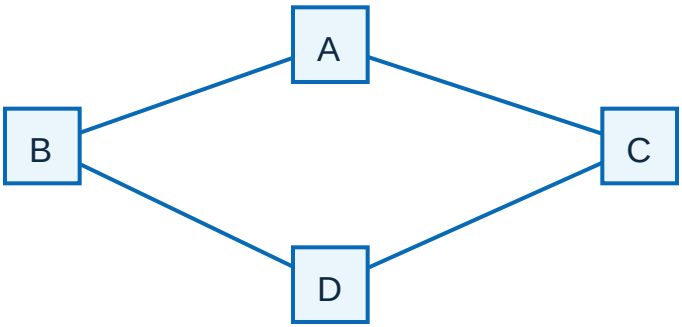
Queue: [A] -> [B, C] -> [D]

Mark a vertex when it enters the queue.

A visited set prevents repeated work.

Depth-first search

Use a stack or recursion to explore one branch first.



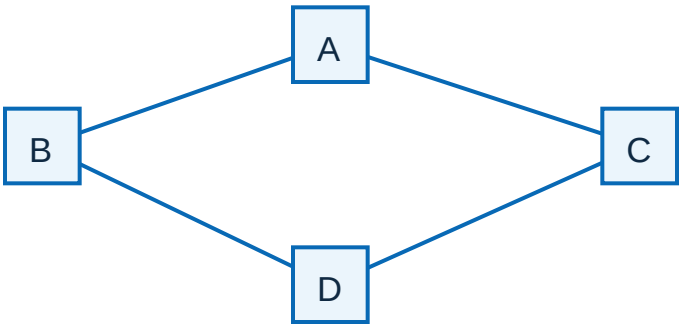
Stack: [A] -> [C, B] -> [C, D]

Track visited vertices before following an edge.

Traversal order depends on neighbor ordering.

Complexity and invariants

With adjacency lists, traversal visits vertices and edges.



Time: O(V + E) for an adjacency-list traversal.

Auxiliary space: O(V) for visited state and frontier.

Disconnected graphs require additional start vertices.